



# Cold shock protein 3 plays a negative role in apple drought tolerance by regulating oxidative stress response

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## ABSTRACT

As RNA chaperones, cold shock proteins (CSPs) are essential for cold adaptation. Although the functions of CSPs in cold response have been demonstrated in several species, the roles of CSPs in response to drought are largely unknown. Here, we demonstrated that *MdCSP3*, a downstream target gene of *MdMYB88* and *MdMYB124*, contributes to drought tolerance in apple (*Malus × domestica*). *MdCSP3* responds to various abiotic stresses, including drought, cold, heat, and salt stress. Compared with non-transgenic apple GL-3, the *MdCSP3* over-expressing plants exhibit significantly lower drought resistance and a reduced capacity for ROS scavenging by the regulation of antioxidant enzymes SOD, CAT, and POD. Additionally, RNA-seq data shows that *MdCSP3* regulates expression of genes involved in oxidative stress response. Taken together, our results demonstrate the functions of *MdCSP3* in apple drought tolerance, and this finding provides a new direction for breeding of drought resistant apple.

## 1. Introduction

Apple (*Malus × domestica* Borkh.) is one of the most popular fruits in the world. Most apple orchards are located in the temperate regions of the northern hemisphere, where they often suffer from various abiotic stresses, including cold, heat, drought, and high salt stress. Among these abiotic stresses, water deficit causes the greatest yield losses (Zhu, 2016). Under drought stress conditions, apple trees use a variety of drought adaptation strategies: activation of antioxidant enzyme activities to scavenge reactive oxygen species (ROS), inhibition of growth vigor to conserve water, regulation of stomatal movement to avoid water loss, greater ABA accumulation, and upregulation of drought-responsive gene expression (Carvalho, 2008). Unraveling the physiological characteristics and molecular mechanisms of the apple drought stress response will establish a basis for enhancing apple drought tolerance and promote the breeding of resistant apple varieties. CSPs (cold shock proteins) were first identified in *Escherichia coli*, where they function as RNA chaperones that destabilize mRNA secondary structures at low temperatures (Jones et al., 1987; Goldstein et al.,

1990). Bacterial CSPs are composed of a single conserved nucleic acid-binding domain called the cold shock domain (CSD). Further study found that the CSD domain contains two conserved sequence motifs, RNP1 and RNP2, that participate in the binding of RNA, ssDNA, and dsDNA in bacteria, animals, and higher plants (Chaikam and Karlson, 2010a; Mihailovich et al., 2010; Wang et al., 2010). Unlike bacterial CSPs, plant CSPs contain both the CSD domain at the N terminus and variable numbers of C-terminal CCHC zinc finger motifs interspersed with glycine-rich regions (Sasaki et al., 2013). Previous studies have shown that the C-terminal zinc fingers and the glycine-rich regions in plant CSPs are important for nucleic acid binding, especially the binding of dsDNA (Karlson et al., 2002; Park et al., 2010).

Wheat CSP1 (WCSP1) was the first characterized plant CSP; it is specifically induced by cold acclimation but does not respond to salt, drought, heat, or ABA treatment (Karlson et al., 2002). Other CSPs that participate in environmental stress responses and plant development have been identified in wheat (*Triticum aestivum*), Arabidopsis (*Arabidopsis thaliana*), rice (*Oryza sativa*), and Chinese cabbage (*Brassica rapa*) (Karlson et al., 2002; Radkova et al., 2014; Kim et al., 2009; Kim et al.,

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2013; Chaikam and Karlson, 2010b; Choi et al., 2015). The four CSP genes identified in the Arabidopsis genome are reported to regulate developmental processes, including flower development, silique size, flowering time, leaf development, and embryo development (Piyada et al., 2013; Nakaminami et al., 2009). In addition to these developmental regulatory functions, the AtCSPs also participate in stress response.

AtCSP1 is induced by cold temperatures, and its overexpression improves water-deficit tolerance (Piyada et al., 2013). In wheat, WCSP1 is induced specifically by cold. The WCSP1 protein binds ssDNA, dsDNA, and RNA homopolymers, and its C-terminal zinc fingers participate specifically in binding to dsDNA (Karlson et al., 2002). However, the rice OsCSPs contrast sharply with WCSP1 in terms of both transcript and protein levels. Transcripts of OsCSP1 and OsCSP2 are upregulated rapidly by cold stress and return quickly to their baseline levels. However, OsCSP proteins are maintained at a constant level for several days after cold treatment (Chaikam and Karlson, 2010b). AtCSP2 is the most abundantly expressed Arabidopsis CSP gene, followed by AtCSP4, with which it is functionally redundant (Sasaki et al., 2013; Nakaminami et al., 2009; Sasaki et al., 2015). The expression levels of CBF (C-repeat binding transcription factor) genes and COR (cold regulated gene) genes are upregulated in *atcsp2 atcsp4* double mutants after cold acclimation and salt stress, indicating that AtCSP2 negatively regulates freezing tolerance and salt stress through a CBF-dependent pathway (Sasaki et al., 2015). AtCSP1 and AtCSP2 are also involved in seed germination and growth under stress conditions. AtCSP3 is the most extensively studied CSP gene in Arabidopsis and plays a positive role in plant cold resistance. Its overexpression enhances plant drought and salt tolerance, in part through the ethylene signaling pathway (Kim et al., 2009; Kim et al., 2013). BrCSP3 is located in chloroplasts and positively regulates seed germination and seedling growth under salt stress and mannitol-induced osmotic stress (Choi et al., 2015). Although CSPs participate in stress responses in various plants, little is known about the function of CSPs in apple. In a previous study, we found that MdMYB88 and MdMYB124 acted as positive regulators of apple cold resistance through a CBF-independent pathway by binding directly to the MdCSP3 promoter during cold stress, and activated MdCSP3 contributes to apple cold resistance (Xie et al., 2018). Here, we show that overexpression of MdCSP3 reduces the drought resistance of apple plants by negatively regulating antioxidant stress response.

## 2. Materials and methods

### 2.1. Plant materials, growth conditions

The plant material used for genetic transformation of apple was GL-3, which was selected from progenies of 'Royal Gala' apple (*M. × domestica* Borkh. cv. Royal Gala) and had high regeneration capacity. Firstly, the tissue cultured GL-3 and transgenic plants were grown in Murashige and Skoog (MS) subculture medium (4.43 g L<sup>-1</sup> MS salts, 30 g L<sup>-1</sup> sucrose, 0.2 mg L<sup>-1</sup> 6-BA, 0.2 mg L<sup>-1</sup> IAA, and 7.5 g L<sup>-1</sup> agar, pH = 5.8) under long-day conditions (16-h-light/8-h-dark) at 25 °C. The plants were rooted in rooting medium (2.22 g L<sup>-1</sup> MS salts, 20 g L<sup>-1</sup> sucrose, 0.5 mg L<sup>-1</sup> IAA, 0.5 mg L<sup>-1</sup> IBA, and 7d.5 g L<sup>-1</sup> agar, pH = 5.8) subsequently. Two months later, MdCSP3 OE plants and GL-3 were transplanted from MS medium to soil (substrate, PINDSTRUP, Denmark) in a growth chamber with 22 °C under 16 h light/8 h dark photoperiods and 60% relative humidity. After 2 months, the plant materials were used drought treatment.

To detect the expression level of MdCSP3 in apple tissues, RNA was extracted from *M. prunifolia* seedlings, followed by qRT-PCR. Two-month-old *M. × domestica* Borkh. cv. Golden Delicious grafted on *M. hupenhensis* were used to detect the expression patterns of MdCSP3 in response to various abiotic stresses. For drought treatment, Golden Delicious plants were irrigated to saturation (0 day) and then irrigation was withheld. Mature leaves were collected at 2, 4, and 8 day for RNA

extraction. For salt stress, 200 mM NaCl was applied. For cold and heat stress, the plants were exposed to 4 °C and 45 °C, respectively.

### 2.2. Generation of transgenic apple plants

The 477-bp coding sequence of MdCSP3 was isolated from the leaves of apple 'Golden delicious' and cloned into the overexpression vector pCambia2300. The constructed MdCSP3-pCambia2300 was transformed into *Agrobacterium* (strain EHA105) for transformation. Leaves of 4-week-old GL-3 were used for *Agrobacterium*-mediated transformation according to previous studies (Dai et al., 2013). Primers for vector construction were shown in Supplemental Table 1.

### 2.3. Drought stress treatment

For drought stress treatment of MdCSP3 transgenic plants and GL-3, the soil volumetric water content (VWC) was measured by TDR (FS6430, USA). Two-month-old GL-3 and MdCSP3 OE transgenic plants were watered to saturation (43–48% VWC) and then withheld irrigation. Mature leaves were collected to examine ion leakage, relative water content (RWC), antioxidant enzymes activities, ROS content, and MDA content when soil VWC reached to 5%. The plants were rewatered when the soil VWC reached to 0%. The survival rate was calculated after rewatering for one week. All the plants were grown in a growth chamber with long-day conditions and appropriate temperature and humidity (25 °C, 50–75% humidity).

### 2.4. Measurement of ion leakage and RWC of leaf

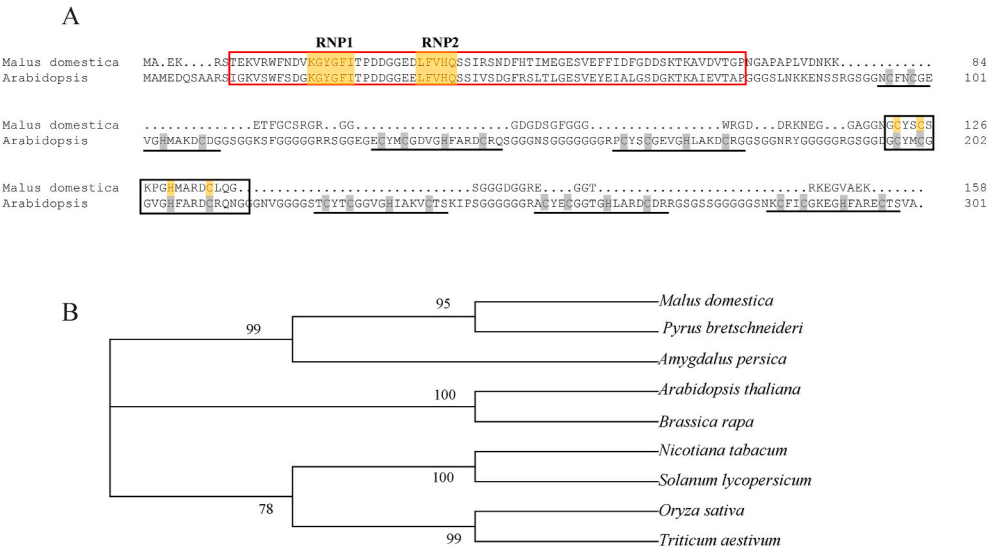
Under drought treatment condition, mature leaves were collected and punched into 8 mm leaf discs with a hole punch. Six leaf discs in each test tube filled with 10 mL deionized H<sub>2</sub>O were used to measure the conductivity (R1). Then the test tube was boiled for 40 min. The conductivity was measured again (R2) when the temperature reduced to the room temperature. The ion leakage was calculated with the formula: (R1/R2) × 100.

For the RWC, plant leaves were collected and weighed quickly (fresh weight, FW), and then immersed in water for 24 h to obtain turgid weight (TW). Finally, the leaves were dried in an oven at 80 °C to obtain constant weight (dry weight, DW). The RWC was calculated as the following formula: RWC(%) = (FW-DW)/(TW-DW) × 100.

### 2.5. Determination of antioxidant enzyme system

The activities of peroxidase (POD) activity, catalase (CAT) activity and superoxide dismutases (SOD) were measured as described previously (Wang et al., 2012). The hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) and superoxide anion content were quantified according to the manuals (Conmin Biotechnology, Suzhou, China). Malondialdehyde (MDA) content was determined as described previously with minor modifications (Wang et al., 2019). Briefly, the leaf samples (0.1 g) were weighed and added with 1 mL extracting buffer (100 mM pH = 7.0 phosphate buffer, 1 mM EDTA, 0.1% TritonX-100, 1% PVP). After mixing on vortex, the mixture was centrifuged at 12,000 rpm at 4 °C for 10 min. The supernatant was used to detect MDA content. The supernatant was mixed with 0.5% (W/V) thiobarbituric acid (TBA) (1/2, V/V) and boiled for 20 min. When the mixture liquid cooled to room temperature, the supernatant was collected and detected with a microplate reader (Infinite M200pro, Tecan, Switzerland) at 532 nm, 600 nm and 450 nm. MDA content (μmol/g) = [6.45 × (A532-A600) - 0.56 × A450]/fresh weight.

For histochemical staining, apple leaves were vacuum-infiltrated for 12 h with 1 mg ml<sup>-1</sup> 3,3'-diaminobenzidine (DAB) or 2 mg ml<sup>-1</sup> nitroterazolum blue chloride (NBT) in 50 mM phosphate buffered saline (PBS, pH = 5.0). Then, the chlorophyll of leaves was removed using 70% ethanol in a water bath of 100 °C. Photos were taken when the chlorophyll was removed clearly.



**Fig. 1.** Sequence and phylogenetic analysis of MdCSP3. (A) Sequence alignment of MdCSP3 from *Malus × domestica* and CSP3 from *Arabidopsis*. The red box indicates a cold shock domain (CSD). The black lines and box indicate C-terminal CCHC zinc finger motifs. (B) Phylogenetic tree of CSPs from different species, including *M. × domestica* (XP\_008340702.1), *Pyrus bretschneideri* (XP\_009363488.1), *Amygdalus persica* (XP\_020423629.1), *Arabidopsis thaliana* (NP\_565427.1), *Brassica rapa* (XP\_009121068.1), *Nicotiana tabacum* (XP\_016483608.1), *Solanum lycopersicum* (XP\_004232496.1), *Oryza sativa* (BAD88072.1), and *Triticum aestivum* (BAD08701.1). Numbers at each fork in the tree indicate the times that gene groups are clustered together in the 1000 bootstrap replicates.

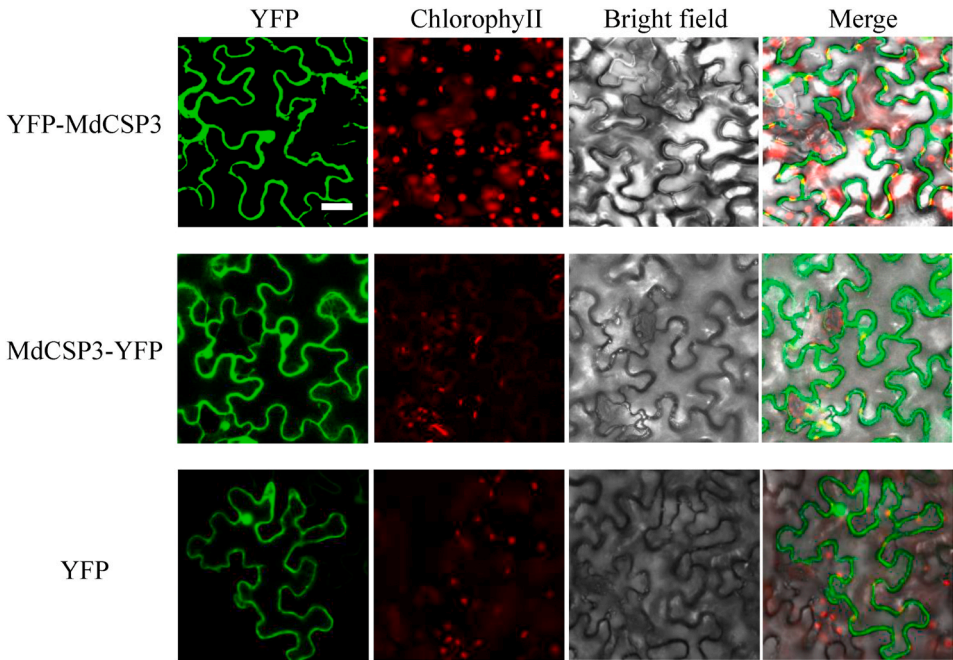
2.6. RNA extraction and qRT-PCR analysis

RNA was extracted from apple leaves using a CTAB method with minor modifications (Chang et al., 1993). During the extraction processing, RNase-free DNaseI (EN0521, Thermo Fisher Scientific, USA) was used to remove DNA. The total mRNA was subjected to reverse transcription using a HiScript® II 1st Strand cDNA Synthesis kit (R212-01, Vazyme Biotech, China). Quantitative reverse transcription PCR (qRT-PCR) was carried out on a Bio-Rad CFX96™ instrument (Bio-Rad, USA) using ChamQ™ SYBR® qPCR Master Mix kit (Q311-02, Vazyme Biotech, China) according to the manufacturer's instructions.

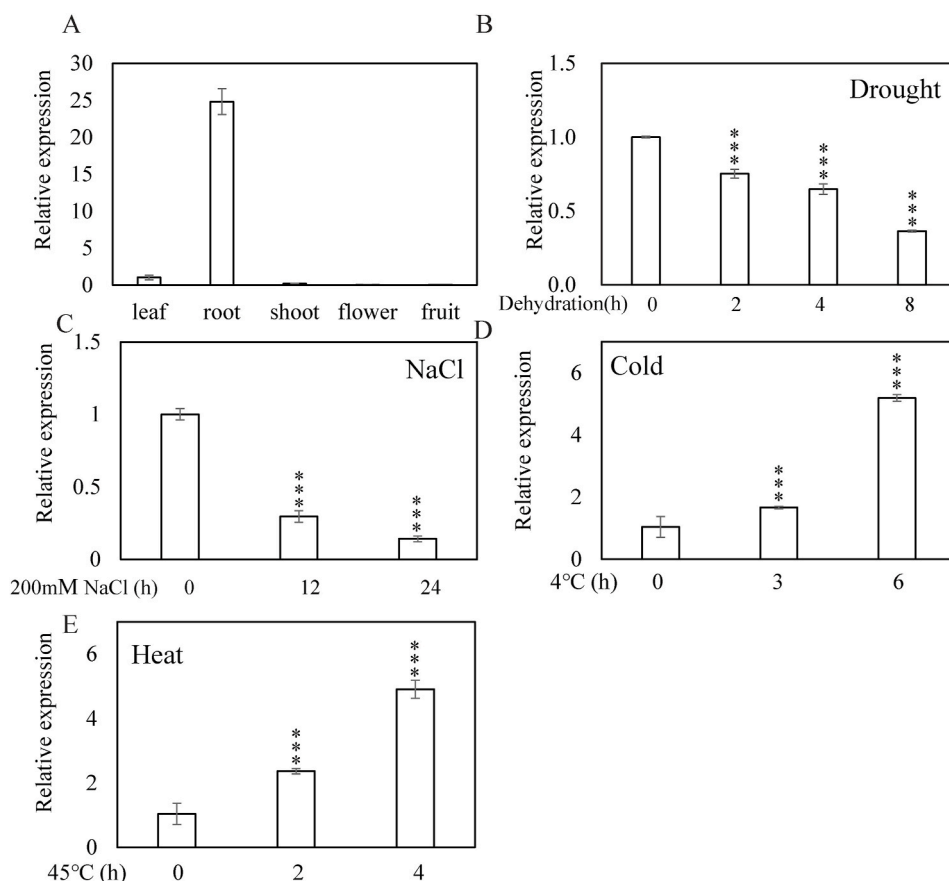
2.7. Transcriptome profiling analysis

Two-month-old MdCSP3 OE and GL-3 plants under control (43–48% VWC) and drought stress (5% VWC) conditions were used as materials.

The RNA was subjected for sequencing using the Illumina HiSeq platform by Novogene (Beijing, China). Raw reads of fastq format sequencing files were firstly processed by trimmomatic v0.39. In this step, clean reads were obtained by removing reads containing adapter, reads containing ploy-N and low-quality reads from raw reads. Then the clean reads were mapping to a latest released *Malus x domestica* genome sequence (GDDH13 version 1.1, [https://iris.angers.inra.fr/gddh13/downloads/GDDH13\\_1-1\\_formatted.fasta.bz2](https://iris.angers.inra.fr/gddh13/downloads/GDDH13_1-1_formatted.fasta.bz2)) by HISAT2 v2.1.0. Following data post-processing including BAM-conversion, sorting and indexing were performed by SAMTools v1.9. Reads of genes in each sample were detected by htseq-count in HTSeq v0.13.5 with apple gene annotation file ([https://iris.angers.inra.fr/gddh13/downloads/gene\\_models\\_20170612.gff3.bz2](https://iris.angers.inra.fr/gddh13/downloads/gene_models_20170612.gff3.bz2)). Gene expression differences were analyzed by R package DESeq2 v1.20.1 with a threshold of adjust *p* value below 0.05 and the absolute value of log2(fold change) greater than 0.5. Length of genes were calculated by GenomicFeatures v1.42.3



**Fig. 2.** Subcellular localization of MdCSP3. YFP-MdCSP3 and MdCSP3-YFP were used for transient transformation in 4-week-old tobacco (*Nicotiana benthamiana*) leaves. Bar = 40 μm.



**Fig. 3.** Expression patterns of *MdCSP3* in different *M. prunifolia* tissues and in response to abiotic stresses. (A) Expression of *MdCSP3* in *M. prunifolia* roots, shoots, flowers, fruits, and leaves. (B)–(E) Expression patterns of *MdCSP3* in response to drought (B), salt (C), cold (D), and heat (E). Data are means  $\pm$  SD ( $n = 3$ ). Mature leaves of two-month-old *M. × domestica* Borkh. cv. Golden Delicious grafted on *M. hupenhensis* were used to detect the expression patterns of *MdCSP3* in response to various abiotic stresses. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by \* $P < 0.05$ , \*\* $P < 0.01$ , or \*\*\* $P < 0.001$ .

and fragments per kilobase of transcript per million fragments mapped (FPKM) values were obtained by Tbttools. Venn diagrams were also plotted by Venn in Tbttools. Heatmaps of gene expression level were plotted by pheatmap v1.0.12. Gene Ontology (GO) annotation and enrichment were analyzed by clusterProfiler v3.18.1.

## 2.8. Subcellular localization assays in *Nicotiana benthamiana* leaves

For subcellular localization, the coding sequence of *MdCSP3* was cloned into the pEarleyGate104 vector to generate YFP-*MdCSP3* fusion protein, and the coding sequence of *MdCSP3* was cloned into the pEarleyGate101 vector to generate *MdCSP3*-YFP fusion protein. Then the resulting vector was transferred into *Agrobacterium* (strain C58C1) and co-injected into 4-week-old tobacco (*Nicotiana tabacum*) leaves with 35S:P19 (P19 is an RNA silencing repressor protein from Tomato bushy stunt virus (Mohammadzadeh et al., 2016)) *Agrobacterium* (strain C58C1). Three days later, the signal of yellow fluorescent protein (YFP) in the co-injected tobacco leaves was detected with a Nikon A1R/A1 confocal microscope (Nikon, Japan).

## 2.9. Statistical analysis

Each experiment was performed in three replicates. The experimental data were analyzed with SPSS 21.0 software. Error bars denote standard deviations. The analysis of the significance level was performed according to Tukey's test at \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ .

## 3. Result

### 3.1. Sequence and phylogenetic analysis of apple *MdCSP3*

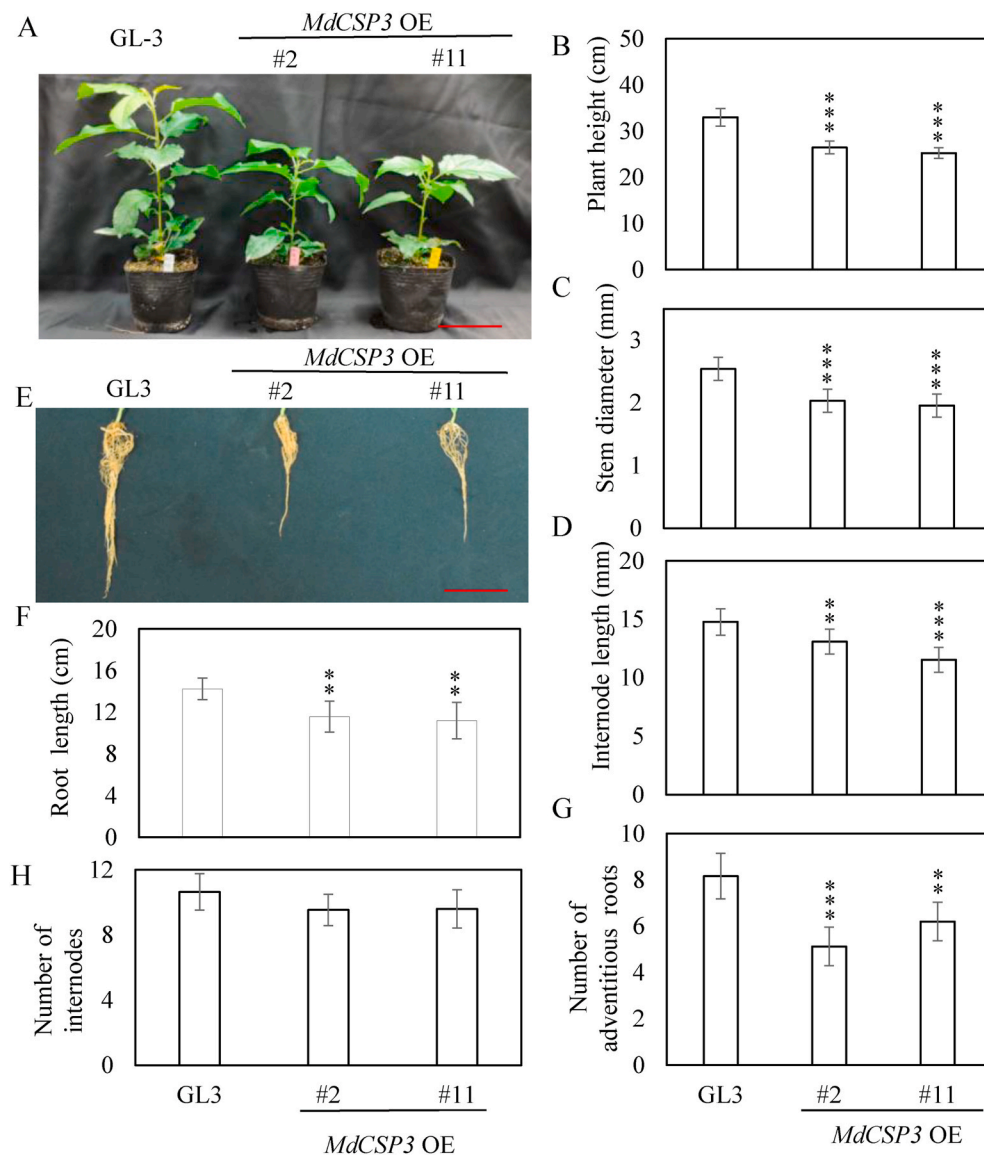
To analyze the function of *MdCSP3* in drought stress response, we

identified the coding sequence of *MdCSP3* in the apple genome (MD16G1153300, GDDH13 V1.1 database). Then we amplified *MdCSP3* from 'Golden delicious'; it comprised 158 amino acids with a predicted molecular mass of 16.61 kDa. Protein sequence analysis with SMART (<http://smart.embl.de/>) identified the conserved structural domains of *MdCSP3*, including a cold shock domain (CSD) in the N-terminal region and a CCHC-type zinc finger domain in the C-terminal region (Fig. 1A). Multiple sequence alignment showed high similarity (31.02%) between the CSD domains of apple *MdCSP3* and Arabidopsis CSP3, but the sequences differed markedly in the C-terminal region. Further phylogenetic analysis revealed high homology among sequences from *M. × domestica* and *Pyrus bretschneideri*, both of which belong to the Rosaceae, followed by *Amygdalus persica*, *Arabidopsis*, and *B. rapa* (Fig. 1B).

### 3.2. Expression patterns of *MdCSP3* in response to various abiotic stresses

We generated YFP-*MdCSP3* and *MdCSP3*-YFP fusion protein for *Agrobacterium*-mediated transient expression in 4-week-old tobacco leaves. After three days, the yellow fluorescence signal was observed in the nucleus and cytoplasm (Fig. 2), implying that these are the main sites of *MdCSP3* function. To further investigate the function of *MdCSP3*, we measured its expression patterns in various apple tissues. Its expression was highest in apple roots, and transcript abundance was low in shoots, flowers, and fruit (Fig. 3A). To determine whether *MdCSP3* participates in the abiotic stress response of apple, we examined its expression in response to different stress conditions. The expression level of *MdCSP3* was reduced when irrigation was withheld (Fig. 3B), and a similar expression pattern was observed following salt treatment (200 mM NaCl). The relative expression of *MdCSP3* fell to 0.14 when the salt treatment lasted for 24 h (Fig. 3C). *MdCSP3* was also upregulated under temperature stress conditions. When plants were exposed to low temperature (4 °C), *MdCSP3* expression gradually increased and reached a





**Fig. 4.** Morphology of *MdCSP3* OE transgenic plants. The morphological characteristics of two-month-old GL-3 and *MdCSP3* OE transgenic lines were analyzed (A), including plant height (B), stem diameter (C), internode length (D), root traits (E), root length (F), number of adventitious roots (G), and number of internodes (H). Bars = 10 cm. Data are means  $\pm$  SD [ $n = 15$  in (B) and (C), 8 in (D), (F), (G), (H)]. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by \* $P < 0.05$ , \*\* $P < 0.01$ , or \*\*\* $P < 0.001$ .

peak of approximately five-fold at 6 h (Fig. 3D). After heat treatment (45 °C), *MdCSP3* was induced approximately five-fold relative to the control condition (Fig. 3E).

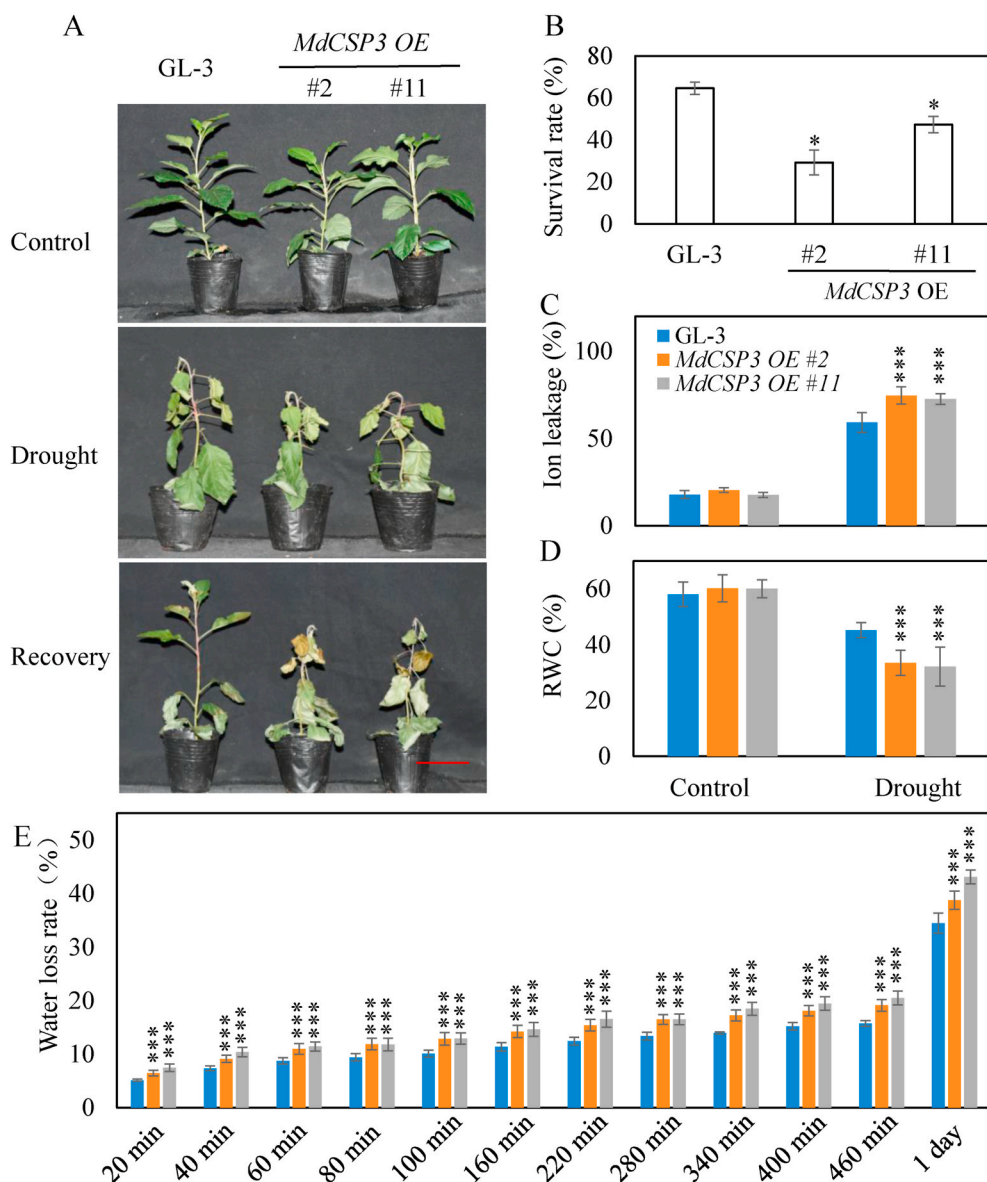
### 3.3. *MdCSP3*-overexpressing transgenic apple plants are dwarfed

To further investigate the role of *MdCSP3*, we transformed 35S::*MdCSP3* into wild-type GL-3 apple plants to obtain *MdCSP3* OE (overexpression) transgenic lines. Using *Agrobacterium*-mediated transformation, we obtained two individual *MdCSP3* OE lines (#2 and #11) whose *MdCSP3* expression level was 4–10 times higher than that of the wild type (Supplemental Fig. 1). Tissue-cultured *MdCSP3* OE and GL-3 plants were transplanted to soil and cultivated in the same plant growth chamber with the same management practices. Two months later, the *MdCSP3* OE transgenic lines were significantly dwarfed relative to GL-3. Statistical tests showed that *MdCSP3* OE plants had lower plant heights, thinner stem diameters, and shorter internode lengths than GL-3 (Fig. 4B–D). However, we did not find significant difference in the internode number between *MdCSP3* OE plants and GL-3 (Fig. 4H). *MdCSP3* OE plants also had weaker root systems, including shorter root lengths and fewer adventitious roots (Fig. 4F and G). All these developmental characteristics suggested that *MdCSP3* overexpression has a

negative effect on plant growth.

### 3.4. Overexpression of *MdCSP3* reduces apple drought tolerance

To clarify the role of *MdCSP3* in the drought stress response, 2-month-old *MdCSP3* OE and GL-3 plants were exposed to drought stress. Before the experiment, all materials were irrigated to saturation (control group), and then irrigation was withheld. When the soil VWC reached to 0%, *MdCSP3* OE plants were completely wilted with dried leaves, whereas GL-3 plants showed less wilting. We then rewetted all the experimental materials and found that 65% GL-3 plants showed recovery after 7 days, whereas 55–70% of the *MdCSP3* OE plants was dead (Fig. 5A). The lower survival rate of *MdCSP3* OE plants suggests their weaker drought tolerance (Fig. 5B). Because electrolyte leakage is an indicator of plasma membrane damage caused by dehydration stress, we next measured the ion leakage of *MdCSP3* OE and GL-3 plants exposed to drought treatment. As shown in Fig. 5C, there was no difference in leaf ion leakage between *MdCSP3* OE and GL-3 plants under control conditions. However, when the soil VWC reached to 5%, *MdCSP3* OE plants suffered more serious damage and showed much higher ion leakage than GL-3 plants (Fig. 5C). In addition to changes in ion leakage, leaf relative water content (RWC) decreased more sharply



**Fig. 5.** Overexpression of *MdCSP3* reduces drought tolerance of transgenic apple plants. (A) *MdCSP3* OE transgenic plants showed weaker drought resistance than GL-3 under drought stress. Bar = 10 cm. (B) Survival rate of *MdCSP3* OE transgenic plants and GL-3. All plants were watered to saturation at the beginning of the experiment (43–48% VWC), then irrigation was withheld. Plants were re-watered when VWC reached 0%, and their survival rate was calculated 7 days later. (C) and (D) Ion leakage (C) and relative water content (RWC, D) of leaves under control (43–48% VWC) and drought (5% VWC) conditions. (E) Water loss rate of detached leaves of GL-3 and *MdCSP3* OE transgenic plants. Data are means  $\pm$  SD [ $n = 3$  in (B), 10 in (C), and (D), 10 in (E)]. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by \* $P < 0.05$ , \*\* $P < 0.01$ , or \*\*\* $P < 0.001$ .

in *MdCSP3* OE plants than in GL-3 plants, decreasing from 60% to 33% compared to 58%–45% in GL-3 (Fig. 5D). We also examined water loss of detached leaves of GL-3 and *MdCSP3* OE plants, and found that the leaves of *MdCSP3* OE plants tend to lose more water under dehydration condition (Fig. 5E). Ion leakage, RWC and water loss data all demonstrated that *MdCSP3* OE transgenic plants experienced greater damage under drought stress, further confirmed their sensitivity to drought stress.

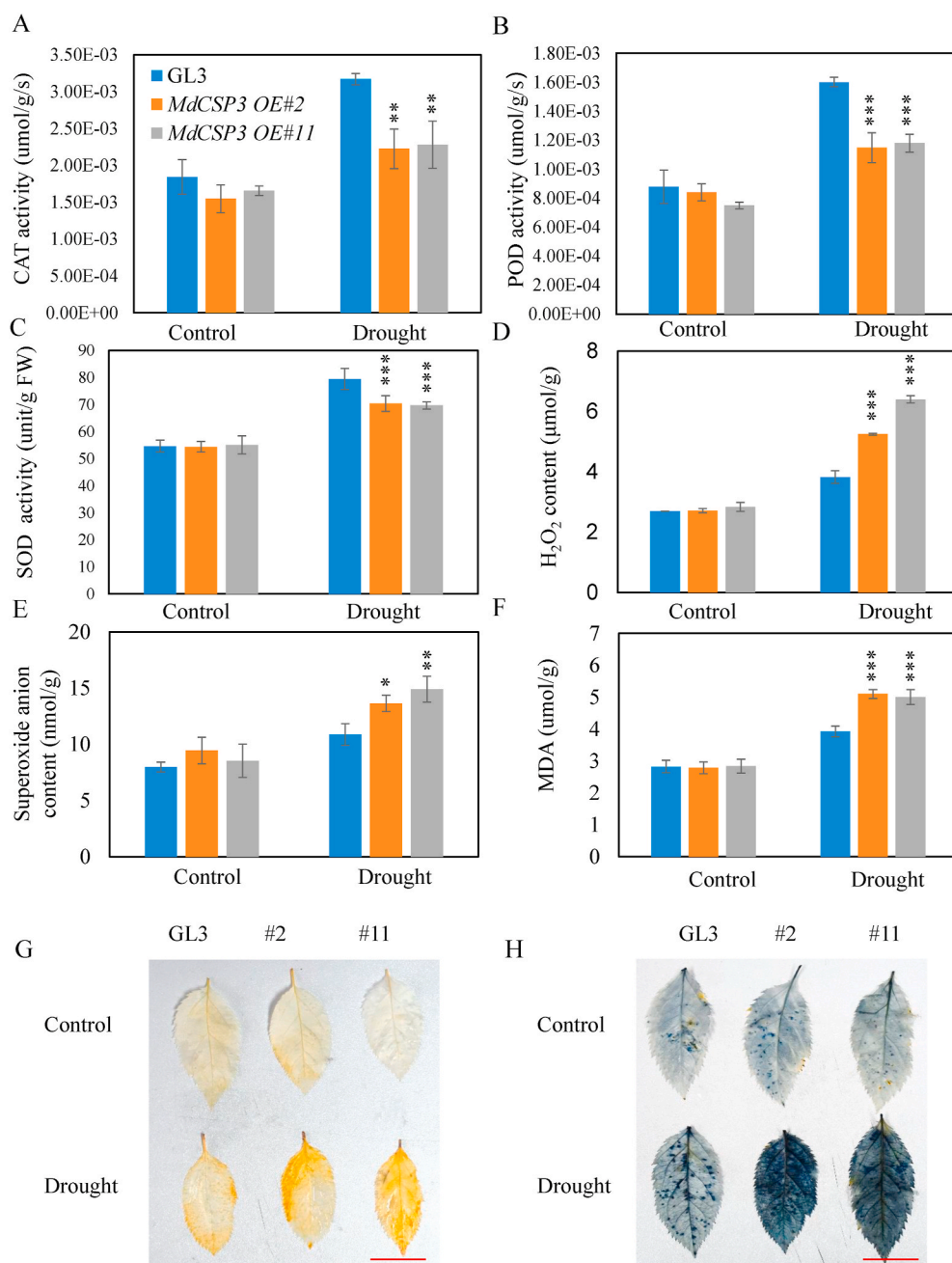
### 3.5. Overexpression of *MdCSP3* reduces the capacity of apple plants to scavenge ROS

When plants are exposed to drought stress, reactive oxygen species (ROS), including  $H_2O_2$  and superoxide anions, accumulate rapidly and cause plasma membrane peroxidation. The damaged plasma membrane further promotes the production of malondialdehyde (MDA) (Ozgur and Gevrek, 2008). We found that the antioxidant capacity of *MdCSP3* OE plants was much lower than that of GL-3 under drought conditions. Although peroxidase (POD), catalase (CAT), and superoxide dismutase (SOD) activities increased in both *MdCSP3* OE and GL-3 plants in response to drought stress, the enzyme activities were significantly

lower in *MdCSP3* OE plants than in GL-3 plants (Fig. 6A–C). With a reduced capacity to scavenge ROS, *MdCSP3* OE plants accumulated more  $H_2O_2$  and superoxide anions under drought stress (Fig. 6D, E, G, H). Consistent with these results, the plasma membrane peroxidation product MDA also accumulated to higher levels in *MdCSP3* OE plants, indicating that their reduced antioxidant capacity led to more serious damage in leaves (Fig. 6F).

### 3.6. Transcriptome profiling of *MdCSP3* OE transgenic plants

To further understand the molecular function of *MdCSP3* in apple drought tolerance, we performed RNA-seq analysis on leaves of drought-treated GL-3 and *MdCSP3* OE plants. The results showed that 7581 genes were upregulated in GL-3 after drought treatment, and 8410 genes were downregulated (Supplemental dataset 1). Relative to GL-3, the expression of 591 genes was upregulated in *MdCSP3* OE plants under drought stress, whereas that of 610 genes was downregulated (Supplemental dataset 2). Among the 591 genes, 293 genes were down-regulated by drought stress; while among the 610 genes, 370 genes were up-regulated by drought (Fig. 7). These data indicate that *MdCSP3* regulates the expression of drought-responsive genes. A Gene Ontology (GO)



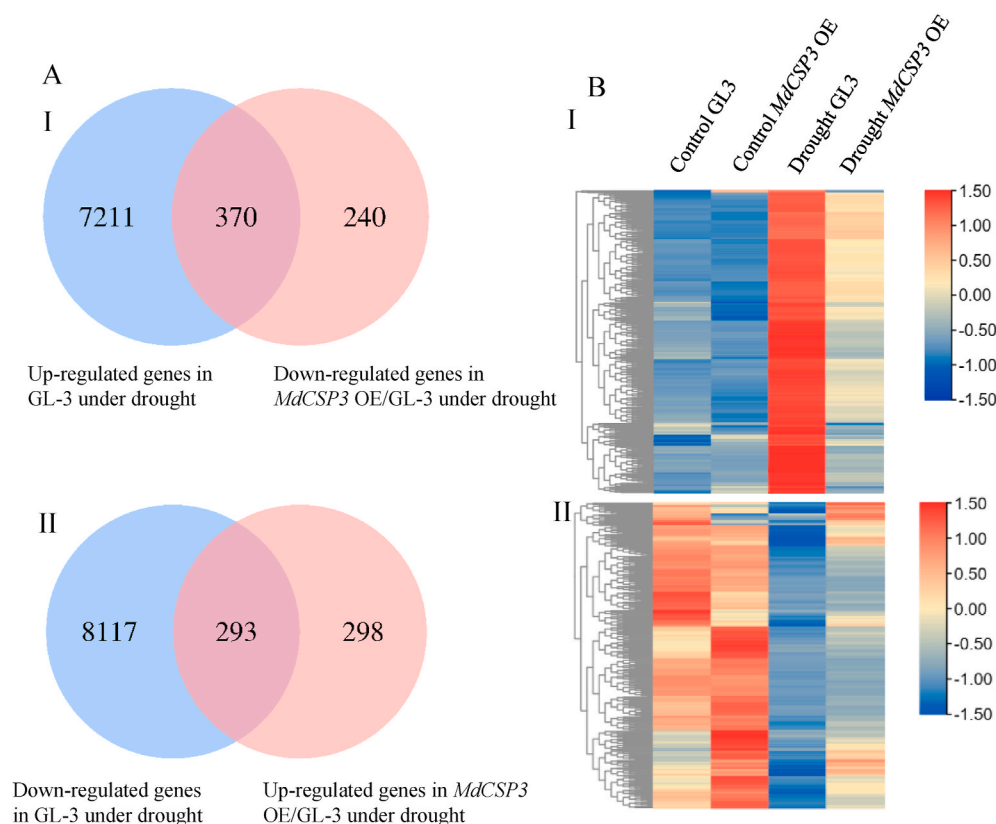
**Fig. 6.** *MdCSP3* OE transgenic apple plants have a reduced capacity for enzymatic scavenging of ROS. (A) Catalase (CAT) activity. (B) Peroxidase (POD) activity. (C) Superoxide dismutase (SOD) activity. (D) H<sub>2</sub>O<sub>2</sub> content. (E) Superoxide anion content. (F) Malondialdehyde (MDA) content. (G) DAB staining showed H<sub>2</sub>O<sub>2</sub> accumulation. (H) NBT staining showed superoxide anion accumulation. Bars = 5 cm. Two-month-old GL-3 and *MdCSP3* OE plants grown with 43–48% VWC condition were used as controls, and treated plants were grown under environment of 5% soil VWC. Data are means  $\pm$  SD [ $n = 9$  in (A)–(C), 10 in (D)–(F), 10 in (G)–(H)]. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by \* $P < 0.05$ , \*\* $P < 0.01$ , or \*\*\* $P < 0.001$ .

enrichment analysis suggested that the differentially expressed genes (DEGs) in *MdCSP3* OE plants were significantly enriched in terms associated with oxidative stress, water deprivation, reactive oxygen species, abiotic stimulus, and environmental stimulus (Supplemental Fig. 2), implying the involvement of *MdCSP3* in oxidative stress response.

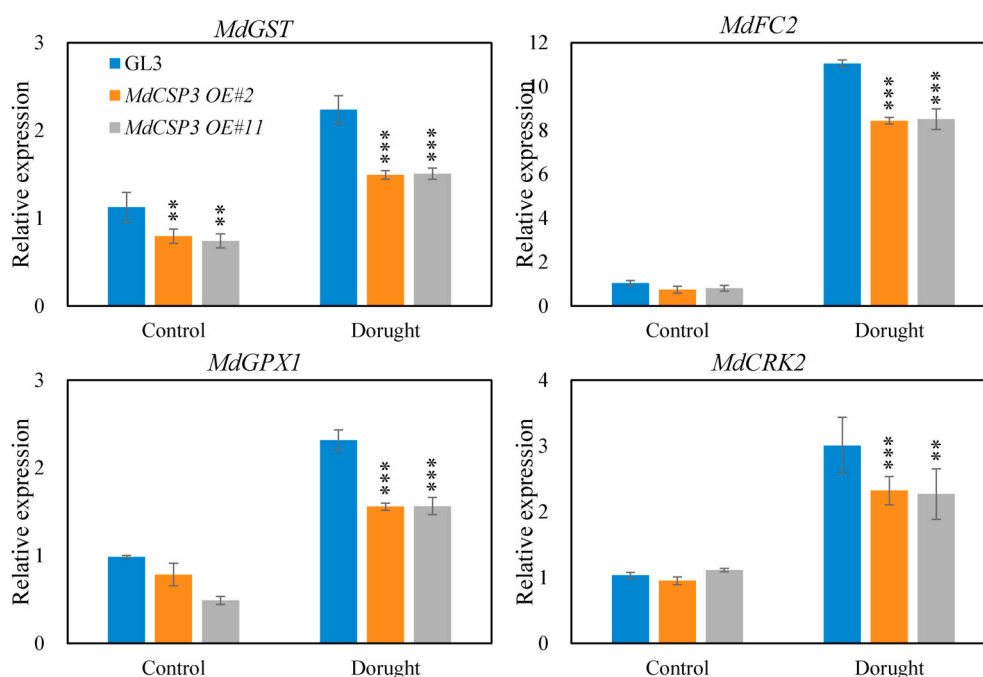
To confirm the reliability of RNA-seq data, qRT-PCR assay was conducted. As shown in Supplemental Fig. 3, expression of *MdRD22*, *MdMYB15*, *MdWRKY40*, and *MdSYT3* were induced by drought in GL-3 and *MdCSP3* OE plants. However, the expression levels of these four genes were lower in *MdCSP3* OE plants compared with GL-3 plants under drought stress condition (Supplemental Fig. 3). These results are consistent with RNA-seq data (Supplemental Fig. 4).

### 3.7. Analysis of oxidative stress-related genes regulated by *MdCSP3* under drought stress

To verify the oxidative stress-related genes regulated by *MdCSP3* under drought treatment, we selected four genes and performed qRT-PCR analysis using GL-3 and *MdCSP3* OE plants under control and drought conditions. The expression patterns of these genes were consistent with those observed in the RNA-seq analysis, confirming the reliability of the RNA-seq data. These four genes, including *MdGST* (glutathione S-transferases), *MdFC2* (ferrochelatases), *MdGPX1* (glutathione peroxidase), and *MdCRK2* (cysteine-rich receptor-like kinases), are known to positively regulate anti-oxidative stress resistance in Arabidopsis or other plants (Miao et al., 2006; Xu et al., 2015; Xu et al., 2016; Woodson et al., 2015; Kimura et al., 2017; Bourdais et al., 2015). They were all upregulated upon drought treatment, but their expression was much higher in GL-3 than in *MdCSP3* OE plants (Fig. 8), which is



**Fig. 7.** Transcriptome profiling analysis. (A) Venn diagram analysis showing the number of DEGs upregulated by the drought treatment and the genes that were down-regulated in *MdCSP3* OE plants under the drought treatment (I); the number of DEGs downregulated by drought, and genes that were upregulated in *MdCSP3* OE plants under the drought treatment (II). (B) Heat-map represents the transcription levels of the DEGs belonging to group I or II as shown in (A). Gene expression level was represented by the FPKM value. DEGs: differentially expressed genes.



**Fig. 8.** Relative expression of oxidative stress-related genes in *MdCSP3* transgenic lines in response to drought. Data are means  $\pm$  SD [ $n = 3$  in (A)–(D)]. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by \* $P < 0.05$ , \*\* $P < 0.01$ , or \*\*\* $P < 0.001$ .

also consistent with RNA-seq data (Fig. 7). These results suggest that *MdCSP3* plays a negative role in drought stress response by regulating the expression of genes related to the oxidative stress response, at least in part.

#### 4. Discussion

Despite the increasing awareness of CSP functions in bacteria and mammals, studies demonstrating the roles of CSP family members in plant stress response and developmental processes are limited to only a



few plant species, including *Arabidopsis*, wheat, rice, and cabbage (Karlsen et al., 2002; Radkova et al., 2014; Kim et al., 2009; Kim et al., 2013; Chaikam and Karlson, 2010b; Choi et al., 2015). Overexpression of *Arabidopsis* CSP3 results in increased drought tolerance, whereas overexpressing MdCSP3 decreases (Fig. 5) (Wang et al., 2012), indicating opposite roles of CSP3 in drought stress response in *Arabidopsis* and apple.

Sequence alignment showed that *Arabidopsis* CSP3 contains a CSD domain with two consensus RNA-binding motifs (RNP1 and RNP2) and a glycine-rich region interspersed with seven CCHC-type zinc finger motifs (Kim et al., 2009; Yang and Dale, 2012). By contrast, apple MdCSP3 contains a CSD domain and only one CCHC-type zinc finger motif. The CSD domain determines the nucleic acid-binding properties and RNA chaperone activity of CSPs. The number and length of the zinc finger glycine-rich domains also contribute to the full activity of CSPs as RNA chaperones and mediators of cold adaptation in *Arabidopsis* (Karlsen et al., 2002; Park et al., 2010). We therefore speculated that the reduced number of zinc finger motifs might endow apple MdCSP3 with functions unlike those of its *Arabidopsis* homolog.

Tissue-specific expression analysis showed elevated expression of *AtCSP3* in meristematic and actively dividing tissues, especially in the shoot apical meristem and reproductive organs, including the inflorescence, open flowers, and siliques (Yang and Dale, 2012). Although these expression patterns imply that *AtCSP3* may function predominantly during reproductive growth, the morphology of *AtCSP3* mutants showed differences mainly in leaf development. The *Arabidopsis* *csp3* mutant exhibits overall reduced seedling size, stunted and orbicular rosette leaves, reduced petiole lengths, and curled leaf blades (Yang and Dale, 2012). However, overexpression of *AtCSP3* did not cause significant morphological differences relative to the wild type (Kim et al., 2009; Kim et al., 2013). Here, we found that MdCSP3 was expressed predominantly in the root (Fig. 3A). However, overexpression of MdCSP3 appeared to impair the root development of transgenic plants, resulting in reduced root length and many fewer adventitious roots (Fig. 4E–G). The dwarf phenotype of MdCSP3 OE plants was in fact similar to the morphology of the *csp3* mutant (Fig. 4A–D) (Wang et al., 2019), further suggesting a unique role for MdCSP3 in apple.

CSP3 was upregulated by cold, salt, drought, and ABA treatment in four-week-old cabbage (Choi et al., 2015), similar to the expression patterns reported for *Arabidopsis* *AtCSP3* (Kim et al., 2009; Kim et al., 2013). By contrast, apple MdCSP3 expression was upregulated by cold and heat stress but inhibited by drought and salt stress (Fig. 3B–E). Previous studies have reported that *AtCSP3* overexpression causes the upregulation of salt-, cold- and drought-responsive genes in *Arabidopsis*, ultimately improving drought and cold resistance (Kim et al., 2009; Kim et al., 2013). Here, overexpression of MdCSP3 in apple reduced the drought tolerance of transgenic plants (Fig. 5). Leaf RWC, ion leakage, and plant survival data all confirmed the reduced drought tolerance of MdCSP3 OE plants. Plants have adapted various strategies to cope with drought stress conditions, including drought avoidance, drought tolerance, drought escape, and drought recovery (Yoshimura et al., 2008; Yue et al., 2005). In apple, drought avoidance and drought tolerance are the major mechanisms of drought stress response. Apple plants avoid water loss through rapid regulation of stomatal movement, leaf rolling, wax accumulation, and enhanced water uptake through a well-developed root system (Geng et al., 2018; Li et al., 2020; Chen et al., 2020; Zhang et al., 2020). Apple plants with strong drought tolerance often maintain normal physiological activities under severe drought conditions by mediating the expression of drought responsive genes, increasing the content of osmoregulatory molecules to maintain cell turgor, and adjusting the activities of antioxidant enzymes to scavenge ROS (Qi et al., 2018). Our study found that the reduced drought tolerance of MdCSP3 OE apple plants was associated with more serious damage to cell membrane integrity and reduced antioxidant enzyme activities for scavenging ROS, including H<sub>2</sub>O<sub>2</sub> and superoxide anions (Fig. 6).

We performed RNA-seq analysis to further explore the mechanisms by which MdCSP3 influences the apple drought stress response. GO enrichment analysis suggested a potential role for MdCSP3 in oxidative stress response. As key factors in plant oxidative stress, H<sub>2</sub>O<sub>2</sub> and superoxide anions both accumulated to higher levels in MdCSP3 OE plants than in GL-3. Apple plants use various methods to cope with oxidative stress, including scavenging ROS through the increased production of antioxidant compounds like ascorbate, glutathione,  $\alpha$ -tocopherol, and carotenoids and through the activation of antioxidant enzymes such as superoxide dismutases (SOD), peroxidases (POD), catalase (CAT), and glutathione reductases (Wrzaczek et al., 2013). Here, we found lower activities of SOD, POD, and CAT in MdCSP3 OE plants than in GL-3 under drought stress (Fig. 6). qRT-PCR results confirmed that many oxidative stress-responsive genes were differentially expressed in MdCSP3 OE and GL-3 plants upon drought treatment. *MdGST*, *MdFC2*, *MdGPX1*, and *MdCRK2* were oxidative response positive regulators and were repressed in MdCSP3 OE plants. We therefore suggest that negative regulation of the apple drought stress response by MdCSP3 depends in part on the oxidative stress response.

Taken together, our results demonstrate that MdCSP3 modulates apple drought stress response through its effects on the expression of oxidative-stress responsive genes, ultimately reducing antioxidant enzyme activities under drought conditions. Overexpression of MdCSP3 also led to a dwarf phenotype and an underdeveloped root system. This weaker growth vigor also resulted in reduced drought resistance.

#### Author contributions

Q.G., X.L. and C.L. designed the project. C.L., N.H., and N.F. performed the experiments. C.L., Z.M., J.H. and F.M. analyzed the data. C. L., X.L. and Q.G. wrote the manuscript. All authors contributed to final manuscript.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.plaphy.2021.10.003>.

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